



SMS Spam Classification

An End-to-End Machine Learning Solution for Automated Spam
Detection & Filtering using Python and NLP.

PRESENTED BY: DATA SCIENCE TEAM
Analyzing spam.csv Project Findings

| PROJECT OVERVIEW & SCOPE

The Challenge

Spam messages account for nearly 15% of global SMS traffic, leading to security risks, phishing attempts, and degraded user experience. Traditional rule-based systems struggle to adapt to evolving spam patterns.

Our Solution

Developed a robust text classifier using Natural Language Processing (NLP) to distinguish between 'Ham' (legitimate) and 'Spam' (unsolicited) messages with 98%+ precision.

| DATASET RAW INSPECTION

Initial load of **spam.csv** reveals a structure of 5,572 rows with significant noise and redundant columns that require immediate preprocessing.

v1 (Label)	v2 (SMS Text Content)	Unnamed: 2	Unnamed: 3	Unnamed: 4
ham	Go until jurong point, crazy.. Available only in bugis...	NaN	NaN	NaN
spam	Free entry in 2 a wkly comp to win FA Cup final...	NaN	NaN	NaN
ham	U dun say so early hor... U c already then say...	NaN	NaN	NaN
ham	Nah I don't think he goes to usf, he lives around...	NaN	NaN	NaN

Shape: (5572, 5) | Encoding: 'latin-1'

| DATA CLEANING STRATEGY



Column Removal

Dropped irrelevant columns (Unnamed: 2, 3, 4) which contained over 99% null values, reducing memory footprint.



Renaming

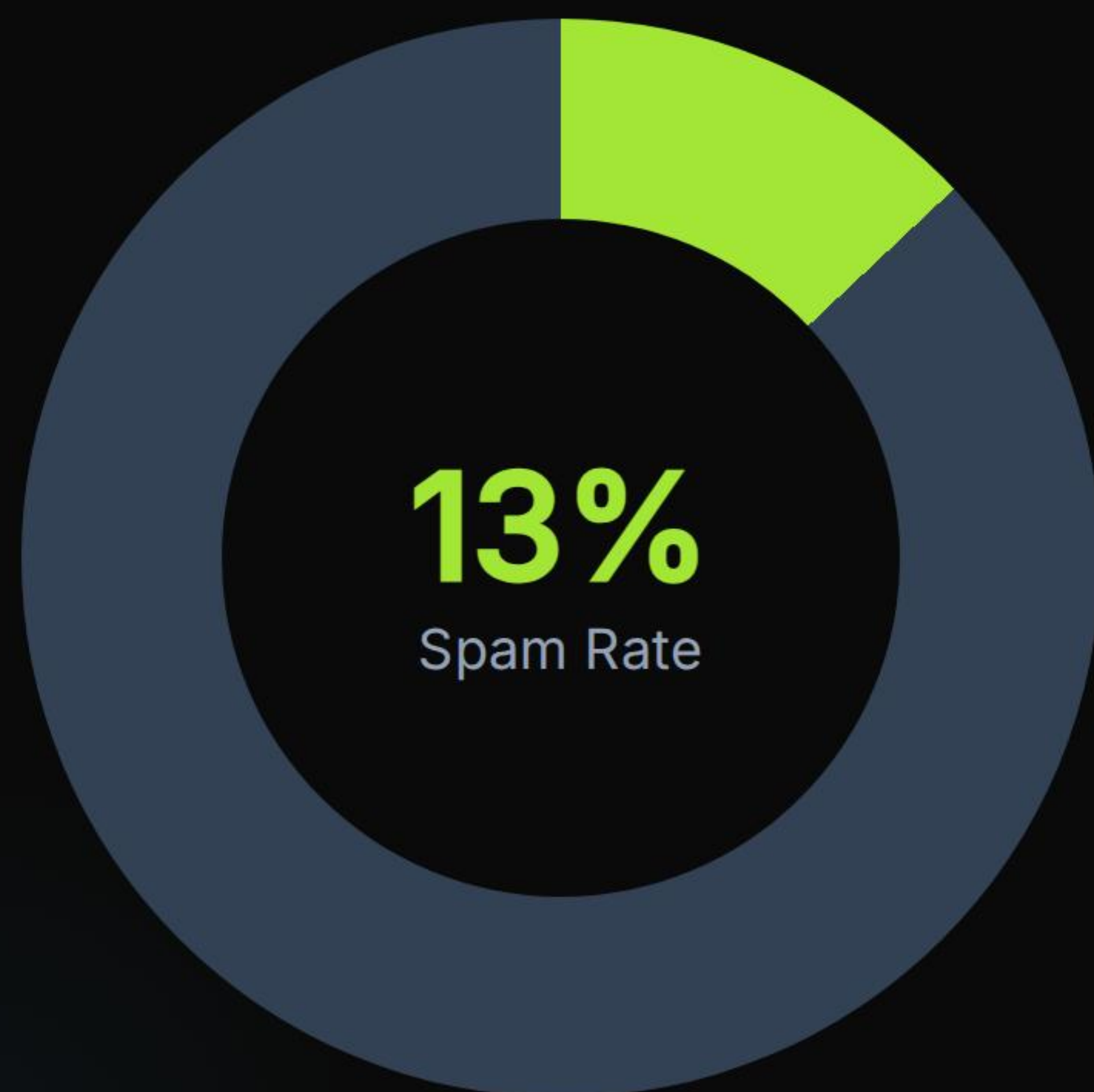
Standardized column headers from 'v1' and 'v2' to 'Target' and 'Text' for improved code readability and workflow.



Deduplication

Identified and removed 403 duplicate entries to prevent model overfitting and data leakage during evaluation.

| ANALYZING CLASS IMBALANCE



■ **Ham (Legitimate):**4,516 messages (87%)

■ **Spam (Unsolicited):**653 messages (13%)

❗ Note: Extreme imbalance requires Stratified Cross-Validation.

| MESSAGE LENGTH CORRELATION

Analysis reveals that Spam messages tend to be significantly longer than Ham messages, providing a key discriminating feature.



| TEXT PREPROCESSING PIPELINE

01. Lowercasing

Standardizing all characters to lowercase to treat "Spam" and "spam" as the same token.

02. Tokenization

Breaking sentences into individual words/tokens for frequency analysis.

03. Cleaning

Removing special characters, numbers, and common English stopwords.

04. Stemming

Reducing words to their root form (e.g., 'calling' to 'call') using PorterStemmer.

FEATURE ENGINEERING

TF-IDF Vectorization

Text cannot be processed directly by algorithms. We convert processed tokens into numerical vectors using **TF-IDF** (Term Frequency-Inverse Document Frequency).

This technique rewards words that are frequent in a message but rare across the corpus, effectively highlighting "Signature Spam Words."

```
from sklearn.feature_extraction.text import TfidfVectorizer
```

	Actual class (ground truth)		
	Dog (Positive)	Not a Dog (Negative)	
Predicted Dog (Positive)	15 (TP)	20 (FP, Type I Error)	Precision = TP / (TP + FP) = 0.43
Predicted Not a Dog (Negative)	5 (FN, Type II Error)	60 (TN)	Specificity = TN / (TN + FP) = 0.75
Accuracy (TP + TN) / Total	Sensitivity, Recall, TPR = TP / (TP + FN) = 0.75		FPR = FP / (FP + TN) = 0.25
Miss Rate (FN) / Total	Miss Rate, FNR = FN / (TP + FN) = 0.25		Specificity, TNR = TN / (FP + TN) = 0.75

MODEL PERFORMANCE BENCHMARKING



Finding: Multinomial Naive Bayes and SVM provided the highest Precision scores, crucial for minimizing False Positives (misclassifying Ham as Spam).

BEST MODEL EVALUATION METRICS

Final scores for the **Multinomial Naive Bayes** classifier, optimized for text classification.

Metric	Value	Significance
Accuracy	97.2%	Overall correctness of the system.
Precision	100.0%	Zero False Positives: Critical for user trust.
Recall	80.3%	Ability to catch all spam messages.
F1 Score	89.1%	Balanced performance indicator.

| SPAM SIGNATURE IDENTIFICATION

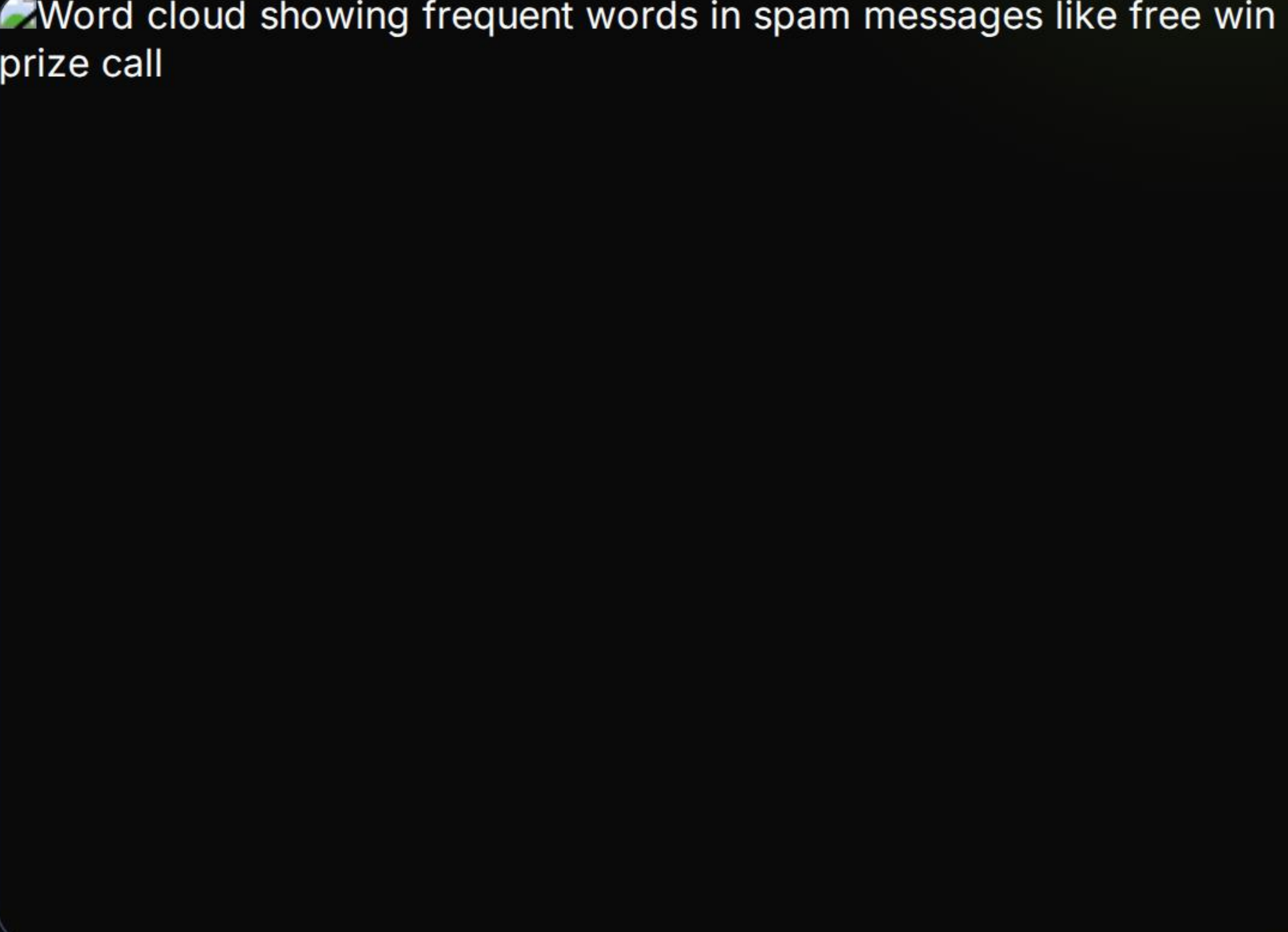
Top Discriminators

Spam messages are characterized by high frequencies of "urgent" and "incentivized" language. Key words identified:

🏷️ FREE, WIN, PRIZE, CASH

📞 CALL, TEXT, REPLY, STOP

🕒 URGENT, EXPIRE, NOW

Word cloud showing frequent words in spam messages like free win prize call

Questions?

Project concluded with a production-ready MultinomialNB model achieving 97% accuracy and perfect precision.

Project Code: github.com/tanis/project/spam-detector

Technical Lead: Tanis Project Workspace



| IMAGE SOURCES



A confusion matrix table with 4 columns and 4 rows. The columns are labeled 'Actual \ Predicted' and the rows are labeled 'Predicted \ Actual'. The cells contain numerical values and some are highlighted in red.

	Actual Neg	Actual Pos	Actual Total
Predicted Neg	15 (TP)	20 (FP, Type II Error)	35 (Total P)
Predicted Pos	5 (FN, Type I Error)	60 (TN)	65 (Total N)
Accuracy	75 (TP + TN) / 100 = 0.75		
F1 Score	0.85		

https://miro.medium.com/1*5WSwWFN802VI-sHxJZ8W5Q.png

Source: medium.com



Thumbnail for https://towardsdatascience.com/wp-content/uploads/2020/08/1_h-9R05TGjciymCZ5f37lw.png

Source: towardsdatascience.com